

Summary of Revisions: FedDil $\rightarrow$ HE-IFD

Manuscript TDSCSI-2026-04-1278 (rejected) $\rightarrow$ resubmission. What changed, and how each reviewer concern is addressed (Ri-W/Q = Reviewer i weakness/question).

Principal change — a method redesign, not a patch. The submitted version trained an encryption-friendly model *under* homomorphic encryption, replacing activations with polynomial approximations and transmitting large encrypted objects over many internal steps; this was the source of the accuracy loss, the ~ 460 GB per-client upload, and the degradation at higher client counts. The resubmission performs *no* training under encryption: each client fine-tunes a small low-rank adapter on a frozen, public, pretrained backbone in plaintext and uploads only the encrypted parameter displacement, and the server’s sole cryptographic operation is one sample-weighted linear sum under multiparty CKKS at *multiplicative depth one*, decrypted by a threshold of clients. Freezing the adapter’s down-projection makes this aggregate *exact* task arithmetic, so the encrypted object is 19 ciphertexts (9.5 MiB), sent once. This resolves the reviewers’ concerns at their root rather than mitigating them.

What changed.

- **No encrypted training:** the only encrypted operation is a depth-one linear combine ($PT \times CT$, $CT + CT$); it reproduces the plaintext aggregate to relative $\ell_2 \approx 10^{-9}$ (zero accuracy cost), with no polynomial activations and no bootstrapping.
- **One-shot, fine-tuning-based:** a single upload and download; learning is local and in plaintext on a frozen pretrained backbone, the setting the encryption constraint admits.
- **Cryptographic, not differentially private:** contributions are protected without perturbation; only the final shared model is revealed, and its residual leakage is measured.
- **New contribution:** encrypted multi-candidate release with post-decryption client-side selection, which also yields a measured defense against a poisoning client at no homomorphic cost.
- **Evaluation and exposition:** language (RoBERTa), vision (ViT-B/16), and a 0.5B language model, with five ablation axes and matched comparators; end-to-end multiparty CKKS measured in Lattigo; prose rewritten plainly, new figures, and the pseudocode replaced by formal equations.

How the rejection points are addressed.

Point	Previous concern	Resolution in the resubmission
R1-W1	Accuracy degrades at higher client counts	No from-scratch encrypted training; accuracy stable across $N = 10\text{--}50$ (DBpedia 0.93/0.90/0.93).
R1-W2	Operator replacement; unquantified accuracy cost	Operator replacement removed; the cryptographic layer is exact ($\ell_2 \approx 10^{-9}$), so its accuracy cost is provably zero.
R1-W3	Upload size / bandwidth (~ 460 GB)	Only the adapter displacement is encrypted: 19 ciphertexts, 9.5 MiB per client, one round.
R1-W4	Empirical privacy; leakage from released model	Membership inference (loss-threshold + LiRA), external and fellow-client adversaries: at or near chance (AUC 0.49–0.59).
R1-W5	No NLP; missing ablations	Four text tasks, two vision datasets, a 0.5B LM, and five ablation axes.
R2-Q1	Comparison with prior work	Comparison table; HE-IFD run at the published setups of DENSE / FedAUXfdp / FedSD2C; new related work on encrypted and DP transfer learning.
R2-Q2	End-to-end CKKS measurement	Full protocol in Lattigo (key-gen, encrypt, aggregate, threshold decrypt); depth one, no bootstrapping; correctness verified.
R2-Q3	Client-side encryption cost	Measured: ≈ 40 ms/client to encrypt; aggregation 76 ms ($N = 10$) to 0.72 s ($N = 100$).
R2-Q4	Malicious / colluding clients	Threat model stated; a measured leave-one-out vote neutralizes a poisoning client; stronger guarantees as future work.
R2-Q5	Post-release privacy across N , α	MIA reported per task; contributions covered cryptographically (server and sub-threshold coalitions see only ciphertexts).
R2-Q6	Plaintext weights, $CT \times PT$, no encrypted optimizer	Exactly the present design: public weights and init in plaintext, only displacements encrypted, server does $CT \times PT$ and $CT + CT$ only, depth one.
R3-1	Participation incentive	Each client obtains a model over the whole label space, including classes it never observed locally.
R3-2	Challenges / contributions; write directly	Restructured around one difficulty (linear averaging of independently trained models fails under skew) and one solution (frozen backbone supplies the shared frame).
R3-3,4	Motivation placement; overview figure	Motivation moved to the introduction; protocol figure redrawn (frozen backbone, shared init, encrypted displacements, depth-one aggregation, threshold decryption).
R3-5,6	Limited comparison; future directions	Comparison broadened and the method explained, not only its outcomes; future directions in a separate scope-and-limitations discussion.